

The Course Map, **charted.**

A four-and-a-half year course plan for BPCH:C at Guelph —
built around the nuclear / energy chemistry track, optimized for
GPA, earnings, and long-term entry into management.

STUDENT

Marcus

PROGRAM

BSCH.BPCH:C

START

Fall 2026

TRACK

Nuclear → Mgmt

OPTIMIZER

Perf · \$\$ · Joy

CONSTRAINT

No 4U Physics

PREPARED FOR

MARCUS — INCOMING FALL 2026

The decision, on one page

Marcus is committing to the **nuclear / energy chemistry** pathway because it offers the highest baseline salary of any BSc-only BPCH exit, a clear technical-to-management progression, and a defined-benefit pension that adds ~25% effective compensation. This document maps every course choice over 4.5 years against that decision.

TRACK

Nuclear → Mgmt

Station-chemist entry. Manager exit.
PEng-eligible by year 4 post-grad.

TARGET SALARY BY 30

\$110–140K

Senior station chemist or supervisor.
Pension adds ~25% effective.

ANCHOR EMPLOYERS

OPG · Bruce · CNL

All three hire BPCH co-ops at
WT2/3 and offer return tracks.

THE ONE PARAGRAPH VERSION

Year 1 is **physics survival** — Marcus has no 4U physics, which locks him into PHYS*1300 then PHYS*1080. The gentler track, but the **70% co-op gate is calculated right after both physics courses finish**, so they must be protected. Years 2–3 the program runs on autopilot — the BPCH co-op stream is already pre-loaded with the analytical chemistry (CHEM*2400) that lands the WT1 nuclear-or-government lab role. Strategy *activates* at Semester 4, when Marcus picks **nuclear-biased restricted electives**: inorganic chemistry (CHEM*3640/3650), environmental analytical, and a deliberate set of **management-seed Liberal Education courses** (econ, business writing, organizational behaviour) that nobody else in his class will think to take.

Year 1 is a physics survival game

Marcus did not take 4U Physics in high school. This single fact dictates the first two semesters and is the highest-stakes risk in the entire 4.5-year plan.

Guelph's physics requirement for BPCH is normally **PHYS*1500** (calculus-based mechanics) in Semester 1. Without 4U physics, that course is closed and Marcus is routed onto the **life-sciences physics stream**:

COURSE	WHAT IT IS	WHY IT'S ACTUALLY THE EASIER PATH
PHYS*1300 Sem 1 · F2026	Fundamentals of Physics — designed specifically for students without 4U physics. Mechanics, energy, waves, electrical circuits, ionizing radiation.	Algebra-based, not calculus-based. Smaller class. Designed for catch-up students. The ionizing radiation unit is a foot in the door for nuclear-track conversations later.
PHYS*1080 Sem 2 · W2027	Physics for Life Sciences. PHYS*1300 satisfies its prerequisite.	Still life-sciences level. Optics, fluids, thermodynamics, modern physics. Easier graders than PHYS*1080's calc-based siblings. And then physics is done forever.

△ WHY THIS IS THE MOST IMPORTANT PAGE IN THE DOCUMENT

The BPCH co-op program requires a **70% cumulative average after Semester 2**. That cutoff falls *immediately after both physics courses*. A bad grade in PHYS*1300 or 1080 doesn't just hurt GPA — it can **eject Marcus from the co-op program entirely**, which collapses the entire earnings strategy (\$50–80K in work-term income, every WT2/3/4 leverage point, the nuclear conversion thesis).

Mitigation: Book Supported Learning Group (SLG) sessions for both physics courses from week 1 — not week 6. Use the lighter Year 1 chemistry load to *bank a GPA cushion* before Semester 3 (CHEM*2060 + 2400 + 2700 + STAT) hits like a freight train.

The hidden upside: **after Semester 2, Marcus never takes another physics course in BPCH**. Two semesters of pain, then he's free. By contrast, students who took 4U physics get routed into PHYS*1500 which is harder and graded less generously. Not having 4U physics accidentally put Marcus on the more forgiving track.

Five years, semester by semester

Every course Marcus will take, with the nuclear-biased choices already filled in. Locked courses are required by the program. Choice courses are where he chooses to lean nuclear. Liberal Ed slots are deliberate management seeds.

■ REQUIRED / LOCKED ■ NUCLEAR-LEANING CHOICE ■ LIBERAL ED · MANAGEMENT SEED

SEM	TERM	COURSES	WHY THIS SET
1	F 2026	CHEM*1040 · General Chem I BIOL*1090 · Molecular & Cellular Biology MATH*1200 · Calculus I PHYS*1300 · Fundamentals of Physics ECON*1050 · Intro Microeconomics	PHYS*1300 is forced. Liberal Ed slot used on Econ — first management-track seed. Marcus needs to read industries (energy markets, regulated utilities) to lead in them; this is where it starts.
2	W 2027	CHEM*1050 · General Chem II MATH*1210 · Calculus II PHYS*1080 · Physics for Life Sci BIOL*1070 · Biodiversity (over 1080) COOP*1100 · 0-cr · Co-op Intro PSYC*1000 · Intro Psychology	BIOL*1070 over 1080 — Biodiversity has historically been the easier-graded biology and protects the 70% gate. PSYC*1000 is the second management seed (people-management starts with understanding people; also reliably high GPA). COOP*1100 is zero credit but required and competitive — take it seriously.
—	S 2027	Summer off. Take a chemistry-adjacent paid job. Read CHEM*2700 (Organic I) textbook chapters 1–4. Do not waste this summer.	
3	F 2027	CHEM*2060 · Structure & Bonding CHEM*2400 · Analytical Chem CHEM*2700 · Organic Chem I STAT*2040 · Statistics	This is the program's hardest semester. Zero flexibility — co-op stream locks all four. CHEM*2400 (Analytical) is the course that lands the WT1 lab role — treat it as a job application. No Liberal Ed slot this semester. Survival mode.
WT 1	W 2028	COOP*1000 · Work Term I. Target: a federal/provincial lab (Health Canada, CNL/AECL, NRC, Public Health Ontario) or a Guelph chemistry faculty research lab. Pay ~\$19–24/hr. The <i>resume signal</i> matters more than the dollars at this stage.	
4	S 2028	CHEM*2480 · Spectroscopy BIOC*2580 · Intro Biochem MICR*2420 · Microbiology BUS*2220 · Organizational Behaviour	Pathway decision committed here. CHEM*2480 (spectroscopy) is the analytical-chemistry workhorse used in nuclear water-chem labs. BUS*2220 is the third management seed — start showing leadership signal to OPG/Bruce HR readers.
5	F 2028	BIOC*3570 · Analytical Biochem CHEM*3430 · Physical Chem I CHEM*3640 · Chem of the Elements I ENVS*3030 · Environmental Chem	CHEM*3640 (inorganic / chemistry of the elements) is the single most nuclear-relevant elective in the entire program — covers transition metals, actinides, coordination chemistry. ENVS*3030 (environmental chem) maps directly to nuclear-station water and effluent chemistry.

SEM	TERM	COURSES	WHY THIS SET
6	W 2029	CHEM*3750 · Organic Chem II CHEM*3650 · Chem of the Elements II CHEM*4020 · Instrumental Analysis MGMT*3020 · Decision Making	CHEM*3650 continues the inorganic sequence — non-negotiable for nuclear. CHEM*4020 (instrumental analysis) is the second analytical pillar. MGMT seed #4 — by Semester 6 Marcus has 4 business/people courses on his transcript, a real differentiator.
WT 2	S 2029	COOP*2000 · Work Term II — target OPG (Pickering / Darlington), Bruce Power, or CNL Chalk River. Radiochem / water-side / station-chemistry. Pay \$26–35/hr. Confirm 8-month placement capacity in the interview. Get verbal "we want you back for Fall" by month 3.	
WT 3	F 2029	COOP*3000 · Work Term III — same employer as WT2. This 8-month block is the entire strategy's leverage point. Own a named project. Negotiate a \$2–4/hr raise at rollover.	
7	W 2030	CHEM*4720 · Organic Reactivity <i>or</i> CHEM*4380 · Adv Inorganic Chem PHYS*2030 · Atomic & Subatomic (<i>elective</i>) MGMT*4000 · Strategic Management <i>free elective</i>	CHEM*4380 (Adv Inorganic) is the 4000-level inorganic capstone — strongest nuclear signal possible on transcript. PHYS*2030 (atomic/subatomic) is optional and gives a "nuclear physics literacy" credit if Marcus's GPA cushion can absorb one more physics. MGMT*4000 is the keystone management seed — strategic management is what nuclear utility managers actually do.
WT 4	S 2030	COOP*4000 · Work Term IV — Conversion. Return to WT2/3 employer with the explicit goal of a written full-time offer before Sem 8 starts. Target \$30–35/hr and \$75–85K base on graduation. Negotiate <i>during</i> WT4.	
8	F 2030	CHEM*4360 · Toxicology of Drugs <i>or other 4000-level</i> 2× restricted electives <i>free elective</i>	Light final semester by design. Marcus should arrive at Sem 8 with a signed full-time offer and use this term to finish strong. Fill remaining 4000-level requirement (≥2.00 credits at 4000-level total across the degree) and protect GPA for his eventual MBA application.

COURSE CODE SANITY

Guelph revises course placement annually (e.g., CHEM*2480 moved to Semester 4 recently; STAT*2040 moved to Semester 3; BIOC*3570 lost its summer offering). **Confirm this map with the BPCH faculty advisor every August** before course selection at bscweb@uoguelph.ca. Treat this document as the strategic spine, not the authoritative course catalog.

Every choice, scored and reasoned

The course map above made several choices. This section opens each one — the options Marcus had, the scoring across his three optimization axes (Performance / Money / Joy), and the reasoning behind the pick.

D-01 · THE PATHWAY COMMITMENT

Decided pre-application · re-confirmed Sem 4

Nuclear vs Pharma vs Discovery

→ Nuclear / energy chemistry, with deliberate management-track overlay.

OPTION	PERF	\$\$	JOY	NOTES
✓ Nuclear	HI	HI	MD	Highest baseline of any BSc-only path. \$80–95K start, \$110–140K senior, \$160–200K+ with PEng/MBA. DB pension adds ~25%. Power dynamics: regulated, hierarchical, technical-to-management is a real ladder, not a fantasy.
✗ Pharma	MD	MD	MD	Broadest, easiest to land, but entry pay is realistically \$50–60K not \$65–80K (current Ontario data — see verification notes). Manager path requires MBA more often than nuclear.
✗ Discovery	LO	LO	HI	Highest ceiling but requires MSc + PhD = 5–6 more years before industry entry. Wrong fit for someone optimizing for near-term money and management.

REASONING

Marcus optimizes for Performance + Money + Enjoyment, with explicit preference for the management ladder. Nuclear is the only pathway that gives all three on the timeline he wants: highest entry pay, clearest internal promotion track, and a workplace structure (utility, regulated, hierarchical) where leadership ambition is rewarded rather than viewed as suspicious. The pension and DB benefits are a quiet additional ~25% on effective comp that the other two paths can't match.

PHYS*1500 vs PHYS*1300

→ PHYS*1300 (Fundamentals of Physics) — the only option available.

REASONING

The calc-based PHYS*1500 has 4U Physics as a hard prerequisite. PHYS*1300 is the course Guelph built specifically for catch-up students. The hidden silver lining: PHYS*1300 → 1080 is the *algebra-based* life-sciences stream, which is genuinely easier-graded than 1500. Marcus's lack of 4U physics accidentally put him on the more forgiving track. **TWO PHYSICS COURSES, BOTH IN YEAR 1, THEN NEVER AGAIN.**

BIOL*1070 vs BIOL*1080

→ BIOL*1070 (Biodiversity).

OPTION	PERF	\$\$	JOY	NOTES
✓ BIOL*1070 Biodiversity	HI	MD	MD	Historically easier graders. Protects the 70% co-op gate. Nuclear pathway doesn't care which biology — biology isn't load-bearing for nuclear at all.
✗ BIOL*1080 Health	MD	MD	MD	More content-heavy. Would have been the pick for the pharma pathway. Useless for nuclear.

REASONING

Biology is a check-the-box requirement for the nuclear track. Optimize purely for GPA protection at the 70% gate. BIOL*1070 wins on Performance and ties everywhere else. If Marcus ever pivots toward pharma or discovery later, he loses nothing — both pathways accept either biology.

Generic "fun" courses vs management seeds

→ Deliberate management-seed sequence — every Liberal Ed slot weaponized.

OPTION	PERF	\$\$	JOY	NOTES
✓ Econ + Psyc + Org-Behaviour + Strategic Mgmt	HI	HI	MD	Reliably high-GPA courses (especially intro Psyc and Econ). Build a "I deliberately prepared for management" story for HR. Differentiate from every other chemistry resume.
x Scatter-shot easy bird courses	MD	LO	HI	Fine for GPA. Wasted positioning. Most BPCH students do this — Marcus shouldn't.
x Hard humanities (history, philosophy)	LO	LO	MD	Risk to GPA, no career upside. Only if he genuinely loves them.

REASONING

Marcus has ~4–5 Liberal Education credits to spend. Most BPCH students treat these as "free GPA points" and pick whatever's easiest. **MARCUS SHOULD TREAT THEM AS A PARALLEL CURRICULUM.** A BPCH graduate with Econ, Org Behaviour, Decision-Making and Strategic Management on the transcript walks into OPG / Bruce Power as *visibly* management-track. None of these courses are GPA killers — Intro Psyc and Intro Econ are reliably 80%+ courses for a strong student. He gets joy axis from genuinely interesting content; he gets money axis from positioning; he gets performance axis from forgiving graders. Triple win.

HIGH IMPACT

Inorganic & environmental vs organic & biochem-heavy

→ CHEM*3640 + CHEM*3650 + ENVS*3030 + CHEM*4020. All-in on inorganic / analytical / environmental.

OPTION	PERF	\$\$	JOY	NOTES
✓ Nuclear stack (inorg + env + instrumental)	MD	HI	MD	This is what makes the WT2 application to OPG/Bruce go from "BPCH student" to "BPCH student visibly trained for our roles." CHEM*3640/3650 = transition metals & actinides. ENVS*3030 = water chemistry. CHEM*4020 = instrumental analysis.
x Pharma stack (organic + medicinal)	MD	MD	MD	Wastes the WT2/3 leverage. Marcus would be a generic pharma applicant against thousands.
x Discovery stack (more BIOG + research project)	LO	LO	MD	Wrong path. Would force grad school.

REASONING

The 3000/4000-level restricted electives are the single most important course-selection moment in the entire degree, because they determine how Marcus's transcript reads to OPG / Bruce Power / CNL HR. Inorganic chemistry is the language of nuclear chemistry — coordination chemistry of uranium and the actinides, transition-metal complexes in coolant systems, lanthanide separations. Environmental chemistry covers the water and effluent monitoring that station chemists do daily. Instrumental analysis (CHEM*4020) is the methods toolkit they'll test him on. **SAME EFFORT AS ANY OTHER ELECTIVE SET, DRAMATICALLY HIGHER SIGNAL.**

Take it or skip it

→ Skip — substitute two lighter 4000-level restricted electives.

OPTION	PERF	\$\$	JOY	NOTES
✓ Skip 4900 → 2× 4000-level electives	HI	HI	MD	Marcus is targeting industry employment, not grad school. CHEM*4900 is a 1.00-credit thesis — heavy GPA and time risk in Sem 7 when WT4 application energy matters more. Two regular electives are lower risk and equally degree-completing.
✗ Take CHEM*4900	MD	MD	HI	Mandatory for discovery / grad school pathway. Optional and risky for nuclear pathway. Only revisit if Marcus pivots toward an MSc.

REASONING

Research projects are the entry ticket for graduate school applications. They are *not* what OPG hires for — OPG hires for analytical chemistry competence, regulatory awareness, and demonstrated reliability (work-term performance grades). The 4900 carries real GPA downside, eats time during the WT4-conversion semester, and produces signal Marcus's target employers don't actually weight. Skip it. *Unless* his thinking shifts toward an MBA later (which is supported by the MGMT seeds, not a research thesis).

What can go wrong, ranked

Six things that could derail the plan, with the realistic likelihood and the move that mitigates each.

RISK	LIKELY	MITIGATION
Year 1 GPA dips below 70%. Physics 1300 or 1080 ambushes him. Marcus is ejected from co-op.	~15–20%	Single biggest risk in the plan. Pre-emptive SLG bookings from week 1. Office hours every week, not just before exams. Treat physics as Marcus's part-time job in Year 1 — 6+ hrs/week minimum. Don't take the heavy Liberal Ed slot in Sem 2; downgrade PSYC choice if needed.
WT1 doesn't materialize. No co-op offer by start of Winter 2028.	~15%	Apply to 25+ jobs, not 5. Cold-email Guelph chemistry faculty for a backup research-lab position. Have it lined up by November 2027.
WT2 employer (OPG/Bruce/CNL) won't extend to WT3. Budget cycle misalignment.	~30%	Confirm 8-month capacity <i>in the WT2 interview</i> , before accepting. Have Plan B identified by month 2: pivot to CNSC (regulator) or another utility in-domain. Do not switch out of nuclear.
Pathway lock-in regret. Marcus tries nuclear at WT2 and hates it.	~20%	WT1 is the rule-out term — pick a non-nuclear lab (e.g. Health Canada or Guelph faculty research) so he tests a different domain cheaply. By WT2 he commits.
No full-time offer at end of WT4. OPG/Bruce headcount tightens.	~25%	Apply broadly in parallel during Sem 7 — CNL, CNSC, OPG's smaller stations, environmental consulting (Hatch, AECOM nuclear divisions). Don't trust conversion alone.
Course re-shuffle. Guelph moves a required course between semesters and breaks the map.	~30% over 5 yrs	This <i>will</i> happen at least once. Confirm with advisor every August. Treat the map as the strategic spine, not the catalog of record.

What to do, when

The actions Marcus needs to take to make the map happen, by semester. Treat this as the one-page he reviews at the start of each term.

- ☐ **SUMMER 2026 · PRE-MOVE-IN Prep physics.** Buy or borrow PHYS*1300 textbook. Do Khan Academy mechanics + waves through May–August. Aim to start Sem 1 already knowing kinematics and Newton's laws. This is the single highest-ROI prep he can do.
- ☐ **F 2026 · SEM 1 Book SLG for PHYS*1300 on day 1.** Don't wait for the first bad quiz. Visit prof office hours every week. Target 75%+ cumulative to hold the GPA buffer.
- ☐ **W 2027 · SEM 2 Take COOP*1100 seriously.** Build the resume Guelph's HR readers expect. 2 mock interviews via Co-op Career Education. Get one Grade-12 chem/bio teacher confirmed as reference. **Hit ≥ 70% cumulative — co-op gate.**
- ☐ **S 2027 · SUMMER OFF Read CHEM*2700 chapters 1–4.** Get a chemistry-adjacent paid job (UHN dishwasher, retail pharmacy, anything). Don't waste this summer doing nothing.
- ☐ **F 2027 · SEM 3 WT1 hunt opens mid-September.** Apply to 25+ postings. Target federal/provincial labs OR a Guelph faculty lab. *Don't target nuclear yet — save that for WT2.* Use WT1 to rule out.
- ☐ **W 2028 · WT1 Execute the foundation term.** Document every technique in a private skills log. Ask manager directly: "What do I need to do to earn a Very Good performance grade?" Get one named reference.
- ☐ **S 2028 · SEM 4 Commit to nuclear.** Email BPCH faculty advisor for nuclear-track recommendations. Attend a Canadian Nuclear Society student event. Confirm CHEM*3640 + 3650 + ENVS*3030 in Sem 5/6.
- ☐ **F 2028 · SEM 5 WT2 applications open.** Target OPG, Bruce Power, CNL Chalk River, SNC Atkins nuclear, Kinectrics. Use WT1 reference aggressively. Ask each interviewer about 8-month placement capacity *before* accepting.
- ☐ **W 2029 · SEM 6 Lock WT2 by end of February.** During onboarding conversation in May, raise WT3 continuation possibility explicitly.
- ☐ **S 2029 · WT2 Earn the rollover.** By month 3 (July), have an explicit conversation with the manager about returning for Fall. Get it in writing before WT2 ends in August.
- ☐ **F 2029 · WT3 Own a named deliverable.** The 8-month block has to produce one specific project he can quote forever — a validated method, a process improvement with measured % gain, a published SOP.
- ☐ **W 2030 · SEM 7 WT4 strategy: convert mode.** Apply back to WT2/3 employer with explicit goal of written full-time offer. Begin researching MBA programs (3–5 yr horizon).
- ☐ **S 2030 · WT4 Convert.** The full-time conversation starts *week one* of WT4, not month four. Target \$30–35/hr WT pay and \$75–85K base on graduation.

☐ **F 2030 · SEM 8 Land softly.** Ideally job-search-free with signed offer in hand. Finish strong: high GPA in 4000-level CHEM and MGMT*4000 — both feed the eventual MBA application.

SOURCES · University of Guelph 2026-27 Undergraduate Calendar (BPCH:C program); Guelph BSc Schedule of Study (bsc.uoguelph.ca/schedule/bpch); Guelph Co-operative Education Salary Guide 2025 (BSc bracket: \$20.46-\$20.62 avg, \$34.68 top, range floor \$17.20 – updated to \$17.60 minimum wage Oct 2025); Ontario nuclear employer career portals (OPG, Bruce Power, CNL). Career salary ranges beyond co-op are industry estimates and will vary. This document supplements but does not replace the BPCH faculty advisor – confirm course placement annually via bscweb@uoguelph.ca. Prepared June 2026.